

**MELBOURNE FERNY CREEK, Australia**WMO: **948720**Lat: **37.8749S**Lon: **145.3500E**Elev: **1684**StdP: **13.82**Time Zone: **10.00 (AUS)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 36.1       | 37.4 | 26.6                          | 21.8 | 58.1 | 30.6 | 26.3 | 51.4 | 32.0                  | 45.7 | 29.6 | 46.2 | 8.2                   | 290  | 0.557 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 16.9                   | 87.1            | 63.2 | 83.0 | 62.5 | 79.5 | 61.4 | 67.4                | 78.3 | 65.6 | 76.8 | 63.9 | 74.9 | 8.8                  | 350  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 64.0                            | 95.0 | 70.5 | 62.0 | 88.4 | 68.9 | 60.2 | 82.8 | 67.3 | 32.8          | 77.3 | 31.3 | 77.0 | 30.0 | 75.2 | 75.4              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 28.4          | 24.3 | 20.5 | DB                         | 33.3 | 95.0               | 1.8 | 4.5                                                | 32.0 | 98.2       | 30.9 | 100.8      | 29.9 | 103.3      | 28.7 | 106.6 | (4) |
| (5)               |      |      | WB                         | 32.2 | 70.9               | 1.9 | 2.0                                                | 30.8 | 72.3       | 29.7 | 73.5       | 28.6 | 74.6       | 27.2 | 76.0  | (5) |

**Monthly Climatic Design Conditions**

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 53.7   | 63.3  | 63.5  | 60.2  | 55.2  | 49.9  | 45.9  | 44.5  | 45.7  | 49.0  | 52.4  | 56.5  | 59.4 |
|                                                                                  | DBStd       | 9.26   | 8.35  | 8.08  | 7.77  | 6.37  | 4.88  | 3.66  | 3.32  | 4.34  | 5.42  | 6.98  | 7.78  | 8.25 |
|                                                                                  | HDD50       | 701    | 1     | 1     | 4     | 19    | 63    | 131   | 174   | 150   | 83    | 51    | 17    | 7    |
|                                                                                  | HDD65       | 4403   | 136   | 117   | 194   | 302   | 469   | 572   | 636   | 600   | 481   | 398   | 278   | 220  |
|                                                                                  | CDD50       | 2059   | 414   | 380   | 320   | 174   | 59    | 9     | 3     | 15    | 52    | 124   | 212   | 297  |
|                                                                                  | CDD65       | 287    | 84    | 76    | 45    | 6     | 0     | 0     | 0     | 0     | 0     | 7     | 23    | 46   |
|                                                                                  | CDH74       | 2376   | 820   | 610   | 296   | 14    | 0     | 0     | 0     | 0     | 1     | 35    | 165   | 435  |
|                                                                                  | CDH80       | 727    | 313   | 179   | 74    | 0     | 0     | 0     | 0     | 0     | 0     | 3     | 24    | 134  |
| Wind                                                                             | WSAvg       | 7.8    | 8.0   | 7.6   | 7.2   | 7.0   | 7.1   | 8.3   | 8.6   | 8.4   | 8.4   | 8.1   | 7.9   | 7.6  |
| Precipitation                                                                    | PrecAvg     | 35.2   | 2.4   | 2.4   | 2.5   | 3.1   | 2.8   | 3.3   | 2.9   | 3.1   | 3.5   | 3.4   | 3.0   | 2.8  |
|                                                                                  | PrecMax     | 46.9   | 4.5   | 7.0   | 4.3   | 7.9   | 5.7   | 6.6   | 5.5   | 4.7   | 6.8   | 6.6   | 6.5   | 6.7  |
|                                                                                  | PrecMin     | 22.0   | 0.5   | 0.2   | 1.3   | 1.0   | 0.7   | 0.9   | 1.2   | 1.0   | 1.3   | 1.1   | 1.1   | 0.3  |
|                                                                                  | PrecStd     | 6.3    | 1.1   | 1.8   | 0.8   | 1.6   | 1.3   | 1.3   | 1.1   | 1.1   | 1.5   | 1.5   | 1.4   | 1.5  |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 95.4  | 90.8  | 87.7  | 75.8  | 65.3  | 57.8  | 55.7  | 62.2  | 69.0  | 78.3  | 83.0  | 91.0 |
|                                                                                  |             | MCWB   | 66.0  | 63.6  | 63.6  | 58.9  | 53.2  | 50.9  | 49.2  | 51.8  | 55.1  | 58.1  | 62.2  | 63.5 |
|                                                                                  | 2%          | DB     | 87.5  | 85.1  | 81.2  | 71.2  | 62.0  | 55.1  | 52.9  | 58.0  | 64.7  | 72.5  | 78.8  | 83.6 |
|                                                                                  |             | MCWB   | 63.7  | 62.8  | 61.6  | 57.7  | 52.9  | 49.9  | 47.2  | 49.7  | 53.6  | 56.1  | 60.1  | 61.8 |
|                                                                                  | 5%          | DB     | 82.7  | 81.2  | 76.7  | 68.0  | 59.8  | 53.1  | 51.0  | 54.9  | 61.4  | 68.5  | 74.6  | 78.7 |
|                                                                                  |             | MCWB   | 63.0  | 62.3  | 60.9  | 55.8  | 52.1  | 48.5  | 46.4  | 47.8  | 51.7  | 54.8  | 58.9  | 60.6 |
|                                                                                  | 10%         | DB     | 77.9  | 77.3  | 72.6  | 65.2  | 57.5  | 51.3  | 49.3  | 52.3  | 58.2  | 64.7  | 70.5  | 74.2 |
|                                                                                  |             | MCWB   | 62.0  | 61.4  | 59.4  | 54.8  | 51.0  | 47.5  | 45.5  | 46.5  | 50.2  | 53.1  | 57.6  | 59.6 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 70.0  | 69.5  | 66.8  | 61.9  | 57.0  | 53.1  | 51.1  | 53.5  | 57.5  | 61.0  | 66.6  | 67.7 |
|                                                                                  |             | MCDB   | 81.8  | 79.1  | 78.1  | 71.4  | 61.6  | 55.3  | 53.0  | 58.4  | 64.0  | 70.1  | 76.5  | 78.2 |
|                                                                                  | 2%          | WB     | 67.8  | 66.9  | 64.4  | 59.4  | 54.7  | 51.0  | 48.7  | 50.9  | 55.1  | 58.5  | 63.0  | 64.7 |
|                                                                                  |             | MCDB   | 79.3  | 77.2  | 75.5  | 66.9  | 59.2  | 53.7  | 50.8  | 56.2  | 62.5  | 68.0  | 73.2  | 76.6 |
|                                                                                  | 5%          | WB     | 65.7  | 65.0  | 62.6  | 57.7  | 53.1  | 49.4  | 47.2  | 48.8  | 52.9  | 56.4  | 61.0  | 62.8 |
|                                                                                  |             | MCDB   | 77.5  | 75.3  | 72.1  | 64.7  | 57.9  | 51.9  | 49.5  | 53.5  | 59.8  | 65.4  | 70.0  | 74.7 |
|                                                                                  | 10%         | WB     | 63.7  | 63.2  | 60.9  | 56.2  | 51.7  | 48.1  | 46.1  | 47.2  | 50.8  | 54.3  | 58.9  | 60.8 |
|                                                                                  |             | MCDB   | 75.1  | 73.1  | 70.3  | 63.2  | 56.2  | 50.5  | 48.4  | 51.2  | 56.9  | 62.8  | 67.9  | 71.6 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 18.0  | 16.9  | 14.6  | 11.0  | 7.4   | 6.1   | 6.2   | 8.2   | 11.2  | 13.9  | 15.6  | 17.0 |
|                                                                                  |             | MCDBR  | 25.2  | 22.4  | 20.6  | 15.1  | 9.7   | 7.8   | 7.9   | 11.5  | 15.1  | 19.4  | 21.7  | 23.8 |
|                                                                                  |             | MCWBR  | 10.2  | 9.3   | 9.0   | 7.3   | 6.3   | 5.7   | 5.5   | 6.6   | 8.2   | 9.8   | 10.7  | 10.6 |
|                                                                                  | 5% WB       | MCDBR  | 21.5  | 20.0  | 17.9  | 13.7  | 8.8   | 6.6   | 6.8   | 10.4  | 14.1  | 17.6  | 18.7  | 20.5 |
|                                                                                  |             | MCWBR  | 11.0  | 10.0  | 8.9   | 7.6   | 6.2   | 5.4   | 5.4   | 6.8   | 8.5   | 10.4  | 10.9  | 10.8 |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.328  | 0.333 | 0.313 | 0.308 | 0.293 | 0.285 | 0.286 | 0.294 | 0.317 | 0.317 | 0.324 | 0.320 |      |
|                                                                                  | taud        | 2.550  | 2.541 | 2.599 | 2.582 | 2.614 | 2.624 | 2.611 | 2.584 | 2.506 | 2.533 | 2.537 | 2.569 |      |
|                                                                                  | Ebn at Noon | 319    | 310   | 304   | 285   | 272   | 266   | 272   | 287   | 297   | 312   | 318   | 323   |      |
|                                                                                  | Edh at Noon | 34     | 33    | 29    | 25    | 21    | 20    | 21    | 25    | 31    | 33    | 34    | 34    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2261   | 1972  | 1574  | 1090  | 713   | 554   | 618   | 876   | 1252  | 1688  | 1999  | 2230  |      |
|                                                                                  | RadStd      | 139    | 106   | 88    | 64    | 44    | 48    | 32    | 61    | 92    | 139   | 116   | 126   |      |

**Historical Trends**

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only            |  | +1.22 | +0.79   | N/A    | N/A     | N/A   | N/A   | N/A         | -328  | +342  | +72   |
| (47) Regional (39 neighbors) |  | +0.90 | N/A     | N/A    | +1.51   | N/A   | N/A   | -31         | -229  | +311  | +108  |

Nomenclature: See separate page